Möbius Inversion

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1 Notation

$\mathbb{N}$ = Natural Numbers

$\mathbb{P}$ = Positive Integers

$\mathbb{Z}$ = Integers

$\mathbb{R}$ = Real Numbers

$\mathbb{C}$ = Complex Numbers

$\mathbf{R}$ = Commutative Unital RING

$[a, b] = \{x \in \mathbb{Z} \,|\, a \leq x \leq b\}$

$[a] = [1, a]$

2 Möbius Function

Consider two functions $f, g \colon \mathbb{N} \longrightarrow \mathbb{C}$ such that

$$g(x) = \sum_{y \leq x} f(y)$$

Then, f can be written in terms of g as:

$$f(x) = \begin{cases} g(x) & x = 0 \\ g(x) - g(x-1) & x \neq 0 \end{cases}$$

This can be proven trivially by expanding the defintion of $g(x)$ in the above relation.

Trivially,

$$g(0) = \sum_{x \leq 0} f(x) = f(0)$$

and,

$$g(x) - g(x-1) = \sum_{y \leq x} f(y) - \sum_{z \leq x-1} f(z) = f(x)$$

Crucially, it doesn't matter, how f and g are defined independently of each other. Just because the relation of $g(x) = \sum_{y \leq x} f(y)$ exists, no matter how else they are originally defined (§ 3), we can go always go from computation of one to another using the same construction.

Also note that, this can be re-written as:

$$f(x) = \sum_{y \leq x} \mu(y, x) \cdot g(y)$$

where we define $\mu : \mathbb{P} \times \mathbb{P} \longrightarrow \{-1, 0, 1\}$ as:

$$\mu(x, y) = \begin{cases} -1 & y = x + 1 \\ 1 & y = x \\ 0 & \textit{otherwise.} \end{cases}$$

With this definition of μ I can now re-write the relation from above as:

$$\boxed{\forall f, g \colon \mathbb{P} \longrightarrow \mathbb{C}, f(x) = \sum_{y \leq x} g(y).g(x) = \sum_{y \leq x} \mu(y, x) \cdot f(y)}$$

Similarly, consider the example where $f, g \colon \mathbb{N} \times \mathbb{N} \longrightarrow \mathbb{C}$ and,

$$g(x, y) = \sum_{z \leq x} \sum_{w \leq y} f(z, w)$$

The relation can be re-written with f in terms of g as:

$$f(x, y) = \begin{cases} g(x, y) & x = 0, y = 0 \\ g(x, y) - g(x-1, y) & x \neq 0, y = 0 \\ g(x, y) - g(x, y-1) & x = 0, y \neq 0 \\ g(x, y) - g(x-1, y) - g(x, y-1) + g(x-1, y-1) & x \neq 0, y \neq 0 \end{cases}$$

Or, if we take some more liberty we can also do:

$$f(x,y) = \begin{cases} 0 & x \notin \mathbb{N} \text{ or } y \notin \mathbb{N} \\ g(x,y) - g(x-1,y) - g(x,y-1) + g(x-1,y-1) & \textit{otherwise.} \end{cases}$$

Similar to the previous function, this one can also be re-written as:

$$f(x,y) = \sum_{x' \leq x} \sum_{y' \leq y} \mu(x,y,x',y') \cdot g(x',y')$$

Where this time we define $\mu : \mathbb{P}^4 \longrightarrow \{-1,0,1\}$ as

$$\mu(x,y,x',y') = \begin{cases} 0 & x' < x-1 \text{ or } y' < y-1 \\ -1 & x' = x-1 \text{ xor } y' = y-1 \\ 1 & \textit{otherwise.} \end{cases}$$

You can take your time to verify the validity of this claim.
Once you do, notice that just like the last time, this time as well all we really need is the relation between f and g. We don't particularly care about the definitions following any rules. Just this relation holding is enough to jump from one to other.

Fun tidbit

If we call μ we defined in the first example μ_1 and the one from the second example μ_2 Then observe that we can rewrite μ_2 in terms of μ_1 as

$$\mu_2(x,y,x',y') = \mu_1(x,x') \cdot \mu_1(y,y')$$

Now for the most important example, $\forall f,g : \mathbb{P} \longrightarrow \mathbb{C}$

$$f(x) = \sum_{d|x} g(d)$$

can be re-written as

$$g(x) = \sum_{d|x} \mu(x,d) \cdot f(d)$$

where $\mu : \mathbb{P}^2 \longrightarrow \{-1,0,1\}$ is defined as

$$\mu(x,y) = \begin{cases} (-1)^{n\left(p \in \text{Primes}:\ p | \frac{x}{y}\right)} & \forall p \in \mathbb{P}, p^2 \nmid \frac{x}{y}. \\ 0 & \textit{otherwise} \end{cases}$$

3 Formal definition and constraints of a Möbius Function